Kush Agarwal

B.Tech CSE (AI), 2027 · Backend and AI Engineering

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SUMMARY

Final-year Computer Science (AI) undergraduate who builds and tests backend systems and AI features end to end: versioned REST APIs in Python and FastAPI, React front ends, PostgreSQL and Redis data layers, and LLM and vector-retrieval components with guardrails. Strong in data structures, algorithms and graph problems (1938 LeetCode contest rating). Looking for an R&D internship building reliable, AI-driven software.

EDUCATION

B.Tech, Computer Science & Engineering (AI) · Noida Institute of Engineering & Technology, Greater Noida 2023 – 2027

CGPA 8.66 · Class XII (CBSE) 86.20% · Class X (CBSE) 94.00%

TECHNICAL SKILLS

Languages	Python, C++, JavaScript, TypeScript, SQL
Backend & APIs	FastAPI, Node.js / Express.js, REST API design, SQLAlchemy, Pydantic
AI / ML	LLM integration (OpenAI, Anthropic APIs), prompt grounding and output validation, FAISS vector retrieval, Scikit-learn
Frontend	React.js, TanStack Query
Data & Infra	PostgreSQL, MySQL, MongoDB, SQLite, Redis, Docker, AWS (EC2, S3), Git / GitHub
Testing	pytest, property-based testing (Hypothesis), Vitest
Core CS	Data structures & algorithms, graph algorithms, OOD, DBMS, operating systems, computer networks

PROJECTS

TaskFlow Pro – Dependency-Aware Workflow & DAG Scheduling Engine · *Python, FastAPI, PostgreSQL, React, TypeScript, LLM APIs, Docker* Sep 2026

- Built a Kanban workflow platform on a DAG engine that rejects circular dependencies with the exact loop, derives Blocked/Ready status and rolls dependents back when a finished task regresses.
- Designed max-based topological schedule propagation that never double-counts delays on converging paths; pushes a delay through 1,000 tasks and 3,000 links in under 1 ms.
- Added LLM dependency suggestions behind five validation gates (schema, id check, de-duplication, quoted-evidence grounding, cycle dry run) with human approval and an offline fallback.
- 61 backend tests including property-based and concurrency tests, run on SQLite and PostgreSQL.
  github.com/MARCIZOS/taskflow-pro

AI-Powered Log Analyzer & Root-Cause Analysis Tool · *Python, FastAPI, PostgreSQL, FAISS, Redis, Docker*

- Engineered a Dockerized, Redis Streams-based log-ingestion pipeline that scales horizontally across consumer groups for high-volume log data.
- Automated root-cause classification with a unit-tested parsing pipeline and FAISS similarity search that surfaces past incidents and fixes instantly.

Distributed Job Scheduler & Task Queue · *Python, FastAPI, React, Redis, PostgreSQL, Docker*

- Built a horizontally scalable scheduler with Redis-backed queues, a PostgreSQL job store and Dockerized worker pools, with retries, dead-letter handling and scheduled runs.
- Achieved 100% job completion at 6 jobs/sec across 3 concurrent workers; React dashboard shows queue depth, throughput and job history live.

ARAIA – Portfolio Diversification Under Macroeconomic Stress · *Python, FastAPI, React, FAISS, Scikit-learn, AWS*

- Built versioned, unit-tested FastAPI APIs for a React risk-analytics platform on AWS (EC2 + S3); a FAISS-powered recommender cut simulated portfolio drawdown by 59%.

EXPERIENCE

Python Developer · LetsGrowMore Jul 2024 – Aug 2024

- Worked with product stakeholders to scope feature requirements and turn them into technical specs, delivered through code review within sprint deadlines.
- Built Python automation scripts (Pandas, Flask) that streamlined recurring data-processing workflows and reduced manual effort.

ACHIEVEMENTS & CERTIFICATIONS

- 540+ DSA problems solved on LeetCode; 1938 contest rating (Knight).
- Top 10 of 120+ teams, HackO'Clock national-level hackathon. Shortlisted for the build round, Contata NCR Hackathon 2026.
- Cisco Networking Academy: Apply AI – Analyze Customer Reviews; Introduction to Modern AI. Infosys Springboard: Generative AI, Python Programming, React.js, Deep Learning.